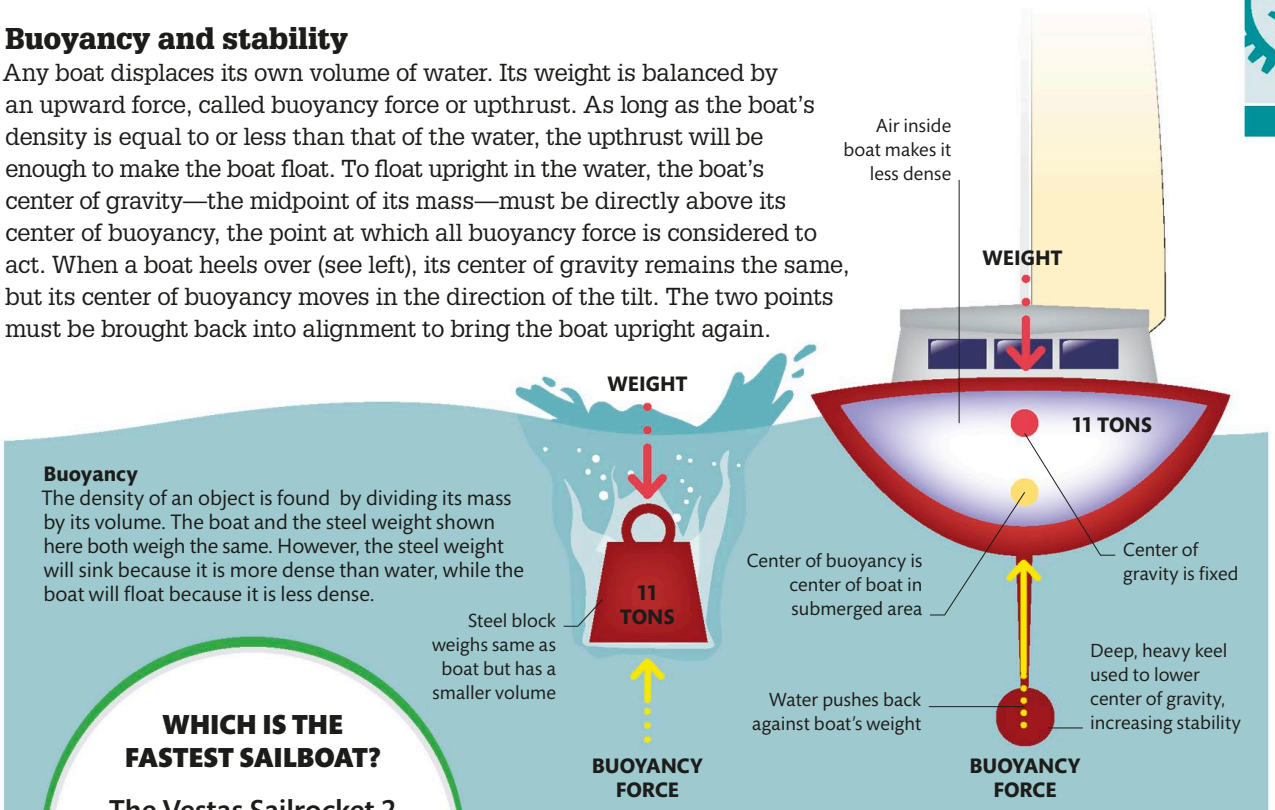
Buoyancy and stability

Any boat displaces its own volume of water. Its weight is balanced by an upward force, called buoyancy force or upthrust. As long as the boat's density is equal to or less than that of the water, the upthrust will be enough to make the boat float. To float upright in the water, the boat's center of gravity—the midpoint of its mass—must be directly above its center of buoyancy, the point at which all buoyancy force is considered to act. When a boat heels over (see left), its center of gravity remains the same, but its center of buoyancy moves in the direction of the tilt. The two points must be brought back into alignment to bring the boat upright again.



Buoyancy
The density of an object is found by dividing its mass by its volume. The boat and the steel weight shown here both weigh the same. However, the steel weight will sink because it is more dense than water, while the boat will float because it is less dense.

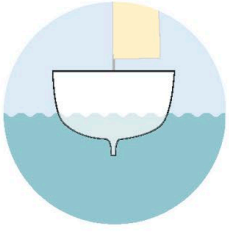
WHICH IS THE FASTEST SAILBOAT?

The Vestas Sailrocket 2 holds the outright world sailing speed record with a speed of 75.2 miles (121.1 km) per hour.

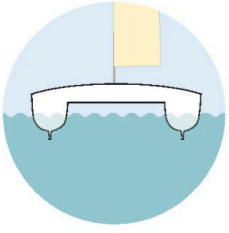
40 DAYS, 23 HOURS, AND 30 MINUTES—THE RECORD-BREAKING TIME TAKEN TO SAIL AROUND THE WORLD

TYPES OF HULLS

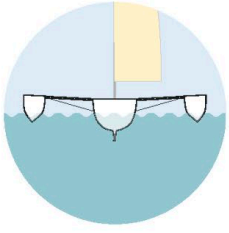
The hull is the main body of a vessel. Sailboats can have one hull (monohulls) or have several hulls (multihulls). Multihulls are often used for racing because they are lighter than monohulls, since they do not require a heavy keel to keep them stable. The most popular multihulls are catamarans and trimarans. Catamarans have two hulls, and trimarans have three.



Monohull
A monohull has a spacious single hull below deck.



Catamaran
Catamarans are wider and so more stable than monohulls.



Trimaran
Trimarans have a main hull with two small outrigger hulls.